

Data Flow Diagram

Date	29 September 2026
Team ID	SWTID-2026-5841
Project Name	FitBuddy – AI Fitness Plan Generator
Maximum Marks	2 Marks

Data Flow Diagram

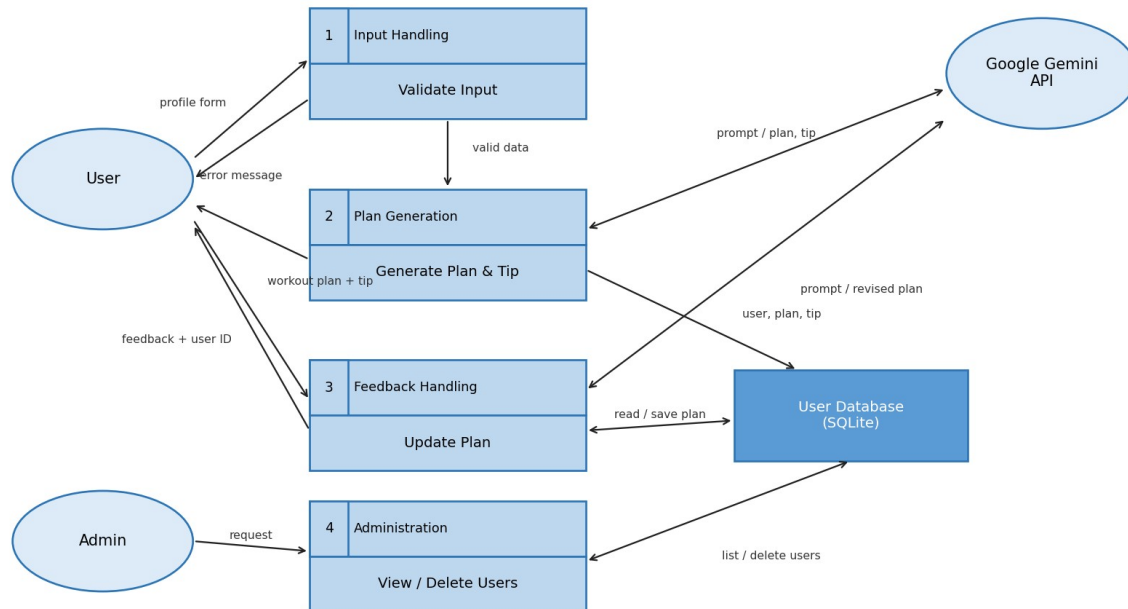
The data flow diagram shows how data moves through FitBuddy between external entities, processes and the data store.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person or system outside the project that sends or receives data (User, Admin, Google Gemini API).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (Validate Input, Generate Plan).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (User Database).
Labeled arrow	Data Flow	The movement of data between entities, processes and data stores.

FitBuddy Data Flow Diagram

FitBuddy - Data Flow Diagram



Step	Flow	Data	Notes
1	User → 1 Input Handling	Profile form: name, user ID, age, weight, goal, intensity	Validated by Pydantic
2	1 → 2 Plan Generation	Valid data; prompt sent to Gemini; plan and tip returned	Google Gemini API
3	2 → User Database	User record, original plan, nutrition tip	SQLite users table
4	User → 3 Feedback Handling	Feedback text and user ID	Latest plan read from database
5	3 → Gemini → Database	Prompt with plan and feedback; revised plan saved	updated_plan column
6	Admin → 4 Administration	View or delete users request	User Database